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Expt.No- 2

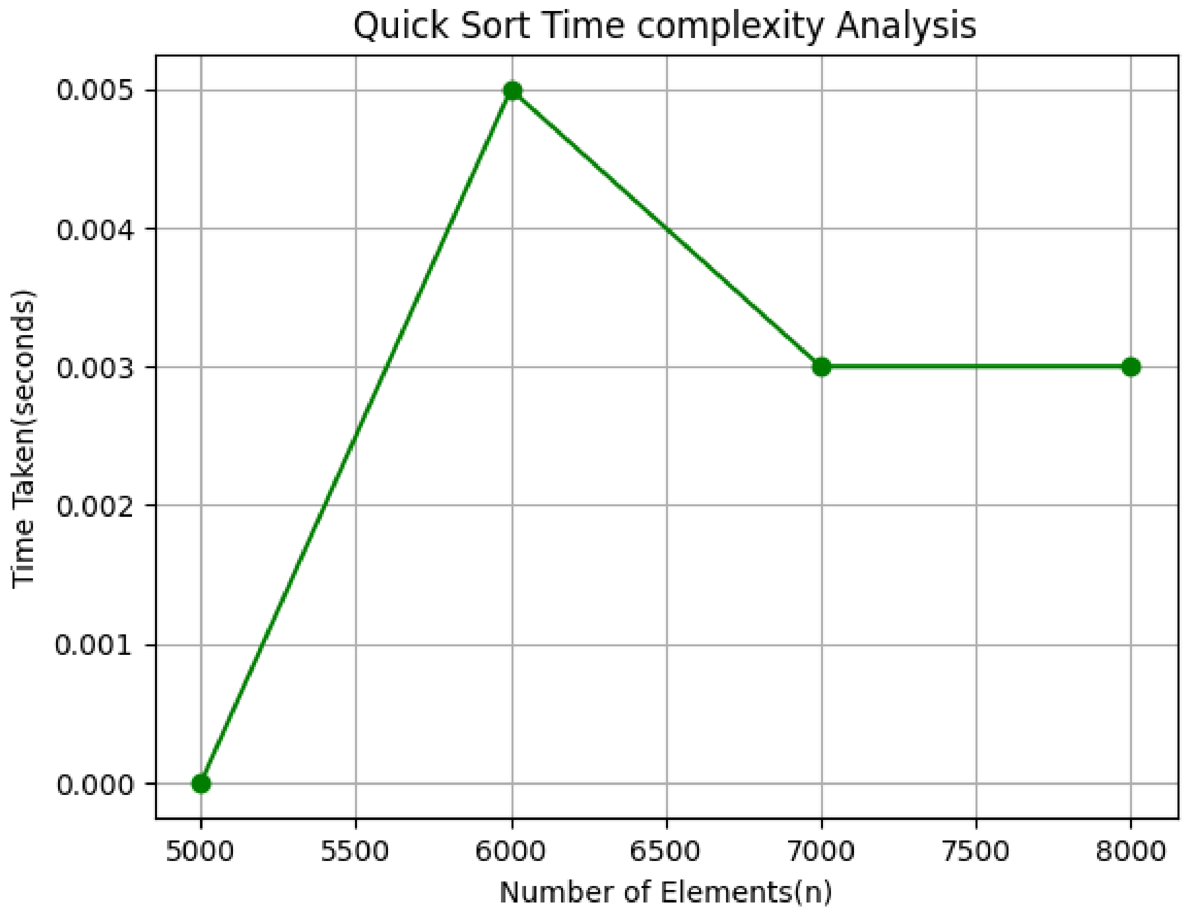
Title - Quick Sort

In [6]:

```
import random
import time
import matplotlib.pyplot as plt

def quick_sort(arr):
    if len(arr)<=1:
        return arr
    pivot=arr[len(arr)//2]
    left=[x for x in arr if x<pivot]
    middle=[x for x in arr if x==pivot]
    right=[x for x in arr if x>pivot]
    return quick_sort(left)+middle+quick_sort(right)

#input:
arr=list(map(int,input("Enter the list of elements separated by space:").split()))
print("Input array:",arr)

sorted_arr=quick_sort(arr)
print("Sorted array:",sorted_arr)

#Measure time taken to sort
start_time=time.time()
sorted_arr=quick_sort(arr)
end_time=time.time()
print("Time taken to sort:",end_time-start_time,"seconds")
```

```
print("==================================================")

# forn>5000
n_values=[5000,6000,7000,8000]
time_values=[]

for n in n_values:
    arr=[random.randint(1,9) for _ in range(n)]

    start_time=time.time()
    sorted_arr=quick_sort(arr)
    end_time=time.time()

    time_taken=end_time-start_time
    print("Time taken to sort",n,"elements:",time_taken,"seconds")
    time_values.append(time_taken)
print("=================================================")

# plotting the graph
plt.plot(n_values,time_values,color='green',marker='o')
plt.xlabel("Number of Elements(n)")
plt.ylabel("Time Taken(seconds)")
plt.title("Quick Sort Time complexity Analysis")
plt.grid()
plt.show()
```

```
Enter the list of elements separated by space:60 40 50 20 10 30 90 70 80
Input array: [60, 40, 50, 20, 10, 30, 90, 70, 80]
Sorted array: [10, 20, 30, 40, 50, 60, 70, 80, 90]
Time taken to sort: 0.0 seconds
==================================================
Time taken to sort 5000 elements: 0.0 seconds
Time taken to sort 6000 elements: 0.004996538162231445 seconds
Time taken to sort 7000 elements: 0.002999544143676758 seconds
Time taken to sort 8000 elements: 0.0029997825622558594 seconds
=================================================
```


Quick Sort Time complexity Analysis
Time Taken(seconds)
0.005
0.004
0.003
0.002
0.001
0.000
5000
5500
6000
6500
7000
7500
8000
Number of Elements(n)